

```
In [1]: # =====
# Notebook setup
# =====

%load_ext autoreload
%autoreload 2

# Control figure size
figsize=(15, 4)

from util import util
import os
import numpy as np
from matplotlib import pyplot as plt
from sklearn.metrics import mean_squared_error

# Load data
data_folder = os.path.join '..', 'data', 'nab'
file_name = os.path.join('realTraffic', 'occupancy_6005.csv')
data, labels, windows = util.load_series(file_name, data_folder)
ddata = data.resample('5min').mean()
ddata = ddata.loc['2015-09-15-06:00': '2015-09-15-23:00']
```

Basic Approaches for Missing Values

Basic Approaches for Missing Values

We will now discuss a few simple approaches to deal with missing values

We will use *partially synthetic data*

- We will focus on specific (and mostly intact) sections of our series
- Then we will remove values artificially
- ...And measure the accuracy of our filling approaches via the Root MSE

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=0}^n (x_i - \hat{x}_i)^2}$$

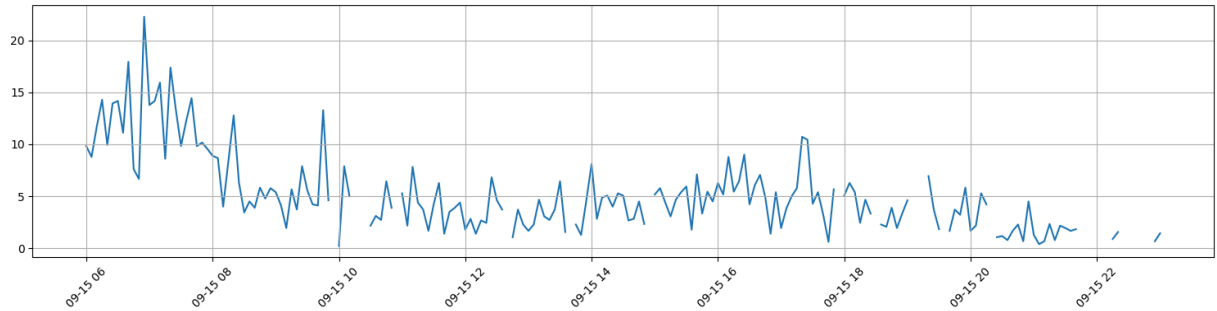
Where $\hat{x}_i$ is a value from the filled series and x_i the ground truth

- $x_i = \hat{x}_i$ if no value is missing
- Hence, any MSE difference is entirely due to missing values

The Benchmark Dataset

Our benchmark dataset will consist of this particular stretch from our traffic series

```
In [18]: util.plot_series(ddata, figsize=figsize)
```



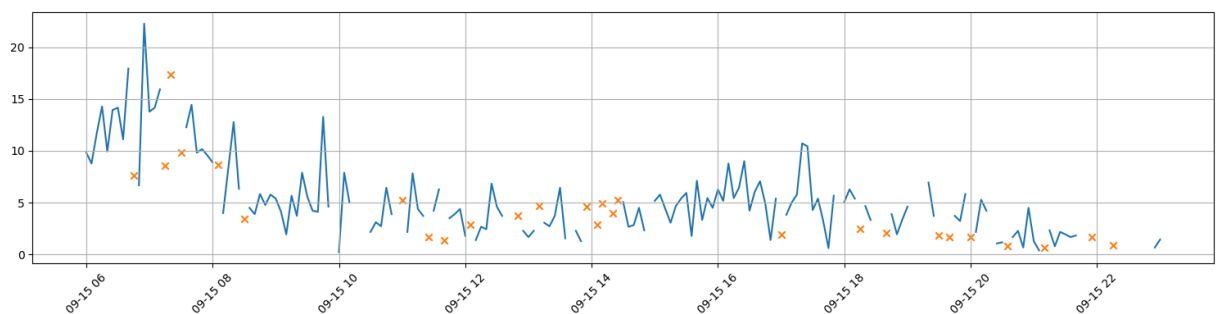
- There are (comparatively) few missing values
- Some of them are isolated, some for contiguous "holes"

The Benchmark Dataset

We now introduce some missing values

...By removing data points at random:

```
In [20]: np.random.seed(42) # seed (to get reproducible results)
mv_idx = np.random.choice(range(len(ddata.index)), size=30, replace=False)
ddata_mv = ddata.copy()
ddata_mv.iloc[mv_idx] = np.nan
util.plot_series(ddata_mv, figsize=figsize)
plt.scatter(ddata.index[mv_idx], ddata.iloc[mv_idx], color='tab:orange', mar
```



- The orange markers represent the value that have been removed

Forward/Backward Filling

The easiest approach for missing values consists in *replicating nearby observations*

- *Forward filling*: propagate forward the last valid observation
- *Backward filling*: propagate backward the next valid observation

An important observation:

- When filling missing values, *we have access to the whole series*
- ...So we can reason *both forward and backward*

Forward/backward filling are simple methods, but they can work well

- Rationale: most time series have a certain "inertia"
- ...I.e.: a strong level of local correlation
- For this reason (e.g.) *the last observation is often a good predictor* for the next one

Forward/Backward Filling on the Benchmark

Forward and backward filling are pre-implemented in pandas

They are available through the `ffill` and `bfill` method:

```
In [21]: nan_mask = ddata['value'].isnull()
ffseries = ddata_mv.ffill()
ffseries[nan_mask] = np.nan # We empty the values that were originally empty
bfseries = ddata_mv.bfill()
bfseries[nan_mask] = np.nan # We empty the values that were originally empty
```

We can check the corresponding RMSE:

```
In [7]: rmse_ff = np.sqrt(mean_squared_error(ddata[~nan_mask], ffseries[~nan_mask]))
rmse_bf = np.sqrt(mean_squared_error(ddata[~nan_mask], bfseries[~nan_mask]))
print(f'RMSE for forward filling: {rmse_ff:.2f}, for backward filling {rmse_bf:.2f}')
```

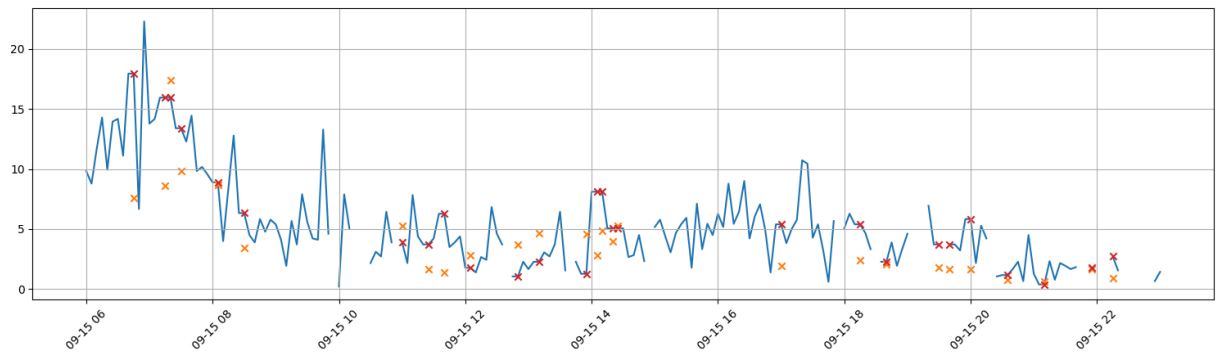
RMSE for forward filling: 1.33, for backward filling 0.87

- In this case backward filling seems to work better
- The results are of course application-dependent

Forward/Backward Filling on the Benchmark

Let's have a close look at the results for *forward filling*

```
In [8]: util.plot_series(ffseries, figsize=figsize)
plt.scatter(ddata.index[mv_idx], ddata.iloc[mv_idx], color='tab:orange', mar
plt.scatter(ddata.index[mv_idx], ffseries.iloc[mv_idx], color='tab:red', mar
```

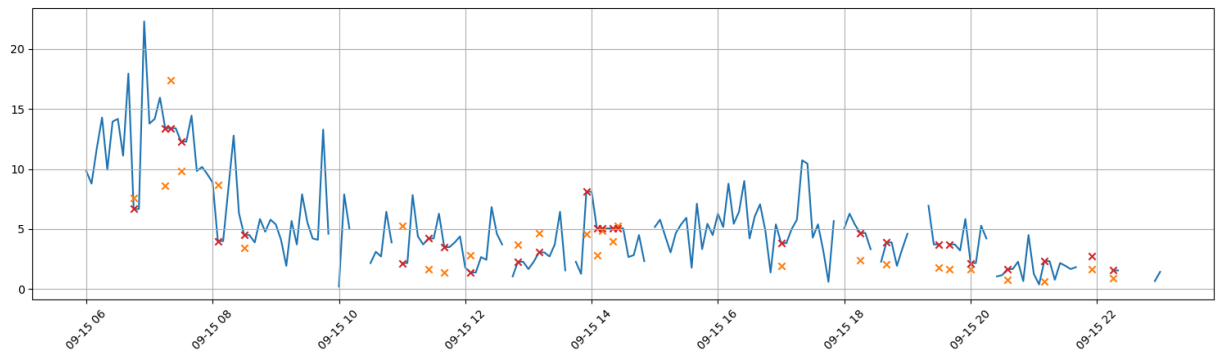


- The red marks represent the values that have been filled

Forward/Backward Filling on the Benchmark

Let's have a close look at the results for *backward filling*

```
In [9]: util.plot_series(bfseries, figsize=figsize)
plt.scatter(ddata.index[mv_idx], ddata.iloc[mv_idx], color='tab:orange', mar
plt.scatter(ddata.index[mv_idx], bfseries.iloc[mv_idx], color='tab:red', mar
```



- Forward/backward filling tend to work well for *low variance* sections
- ...And conversely work worse for *high variance* sections

(Geometric) Interpolation

A few more options are available via the `interpolate` method

`DataFrame/Series.interpolate(method='linear', ...)`

The `method` parameter determines how NaNs are filled:

- "linear" uses a linear interpolation, assuming uniformly spaced samples
- "time" uses a linear interpolation, but supports non-uniformly spaced samples
- "nearest" uses the closest value
- "polynomial" uses a polynomial interpolation
- Even "ffill" and "bfill" are available

Both "polynomial" and "spline" require to specify the additional parameter `order`

- E.g. `df.interpolate(method='polynomial', order='3')`

(Geometric) Interpolation

Let us check the performance of some approaches

```
In [10]: args = [{'method': 'linear'}, {'method': 'time'}, {'method': 'nearest'},  
                {'method': 'polynomial', 'order': 2}, {'method': 'spline', 'order': 4}]  
filling_res = {}  
for a in args:  
    filling_res[a['method']] = ddata_mv.interpolate(**a)  
    filling_res[a['method']][nan_mask] = np.nan  
    rmse = np.sqrt(mean_squared_error(ddata[~nan_mask], filling_res[a['method']]))  
    print(f'RMSE for {a["method"]}: {rmse:.2f}')
```

RMSE for linear: 0.98

RMSE for time: 0.98

RMSE for nearest: 1.35

RMSE for polynomial: 1.10

RMSE for spline: 1.03

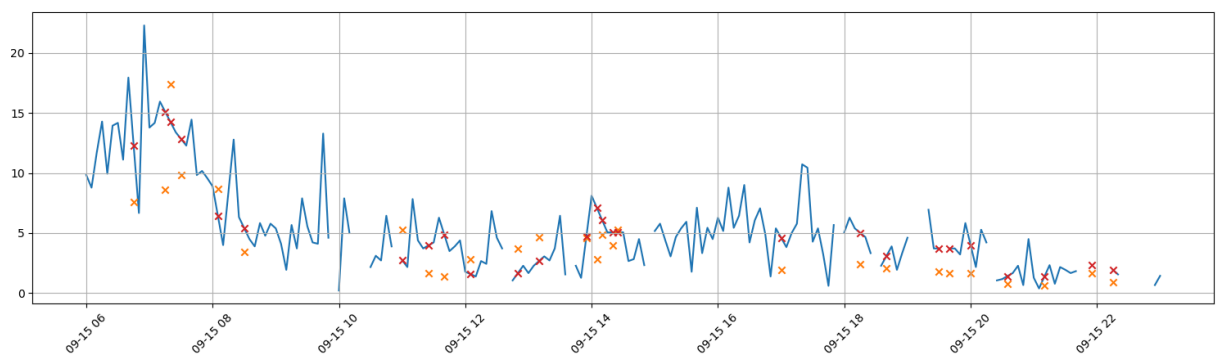
- "linear" and "time" are equivalent (we have uniformly-spaced samples)
- "polynomial" is the most complex, and in this case also the worst

All perform worse than backward filling (at least in this case)!

Linear Filling on the Benchmark

Let's have a close look at the results for *linear filling*

```
In [11]: util.plot_series(filling_res['linear'], figsize=figsize)  
plt.scatter(ddata.index[mv_idx], ddata.iloc[mv_idx], color='tab:orange', mar  
plt.scatter(ddata.index[mv_idx], filling_res['linear'].iloc[mv_idx], color='
```

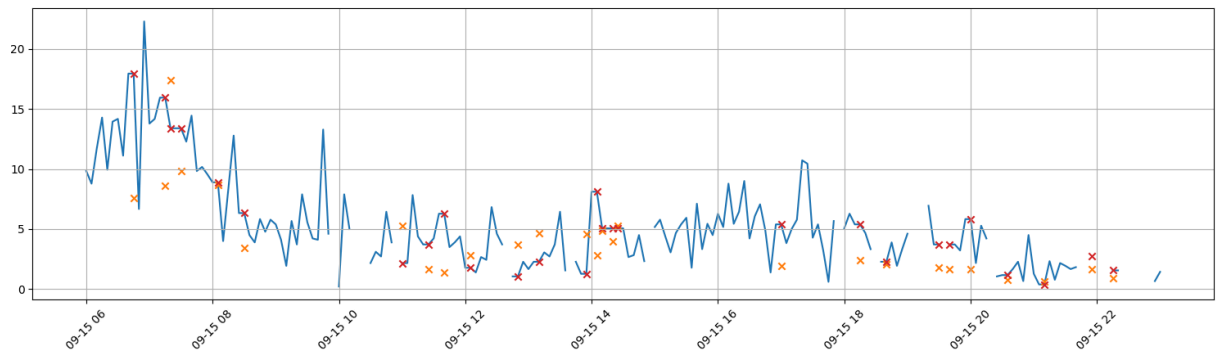


- Linear filling works well for series with *slower dynamics*
- ...But does not work well for series that *faster dynamics*

Nearest Filling on the Benchmark

Let's have a close look at the results for *nearest filling*

```
In [12]: util.plot_series(filling_res['nearest'], figsize=figsize)
plt.scatter(ddata.index[mv_idx], ddata.iloc[mv_idx], color='tab:orange', mar
plt.scatter(ddata.index[mv_idx], filling_res['nearest'].iloc[mv_idx], color=
```

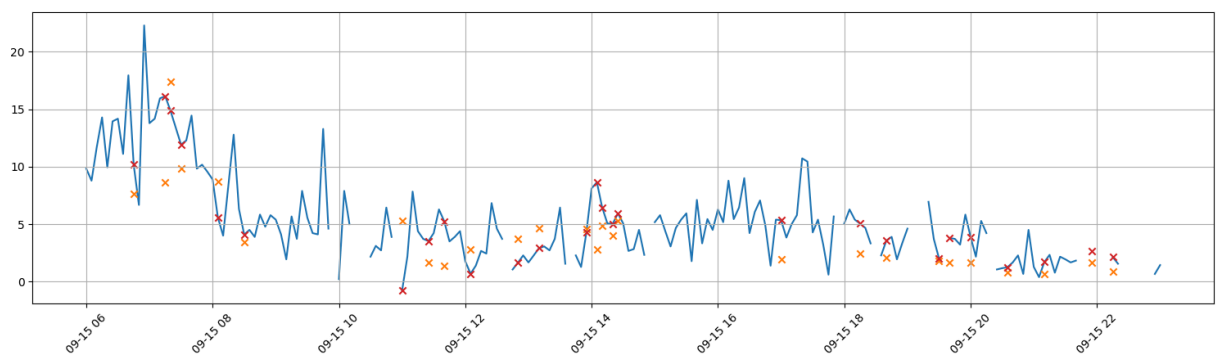


- Nearest filling is a *compromise between forward and backward filling*
- ...And in our case its performance is the average of the two

Polynomial Filling on the Benchmark

Let's have a close look at the results for *polynomial filling*

```
In [13]: util.plot_series(filling_res['polynomial'], figsize=figsize)
plt.scatter(ddata.index[mv_idx], ddata.iloc[mv_idx], color='tab:orange', mar
plt.scatter(ddata.index[mv_idx], filling_res['polynomial'].iloc[mv_idx], col
```

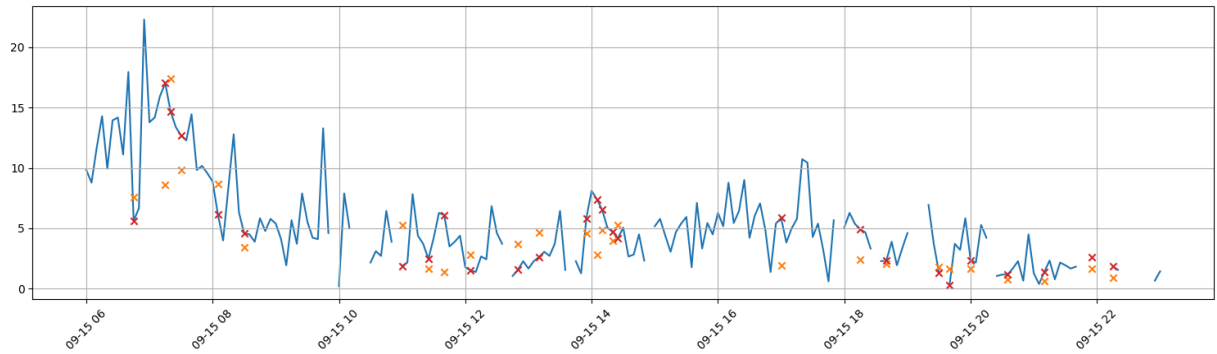


- Polynomial interpolation relies on nearby values to fit a polynomial
- *High-order polinomial often vary too much* and work less well

Spline Filling on the Benchmark

Let's have a close look at the results for *spline filling*

```
In [14]: util.plot_series(filling_res['spline'], figsize=figsize)
plt.scatter(ddata.index[mv_idx], ddata.iloc[mv_idx], color='tab:orange', mar
plt.scatter(ddata.index[mv_idx], filling_res['spline'].iloc[mv_idx], color='
```



- Spline interpolation relies on [piecewise polynomial curves](#))
- ...And it's often *more robust than polynomial interpolation*

How do we choose which method is best? Is the (R)MSE enough?